Model intrinsic termination via its sequences and determine transient kinetics

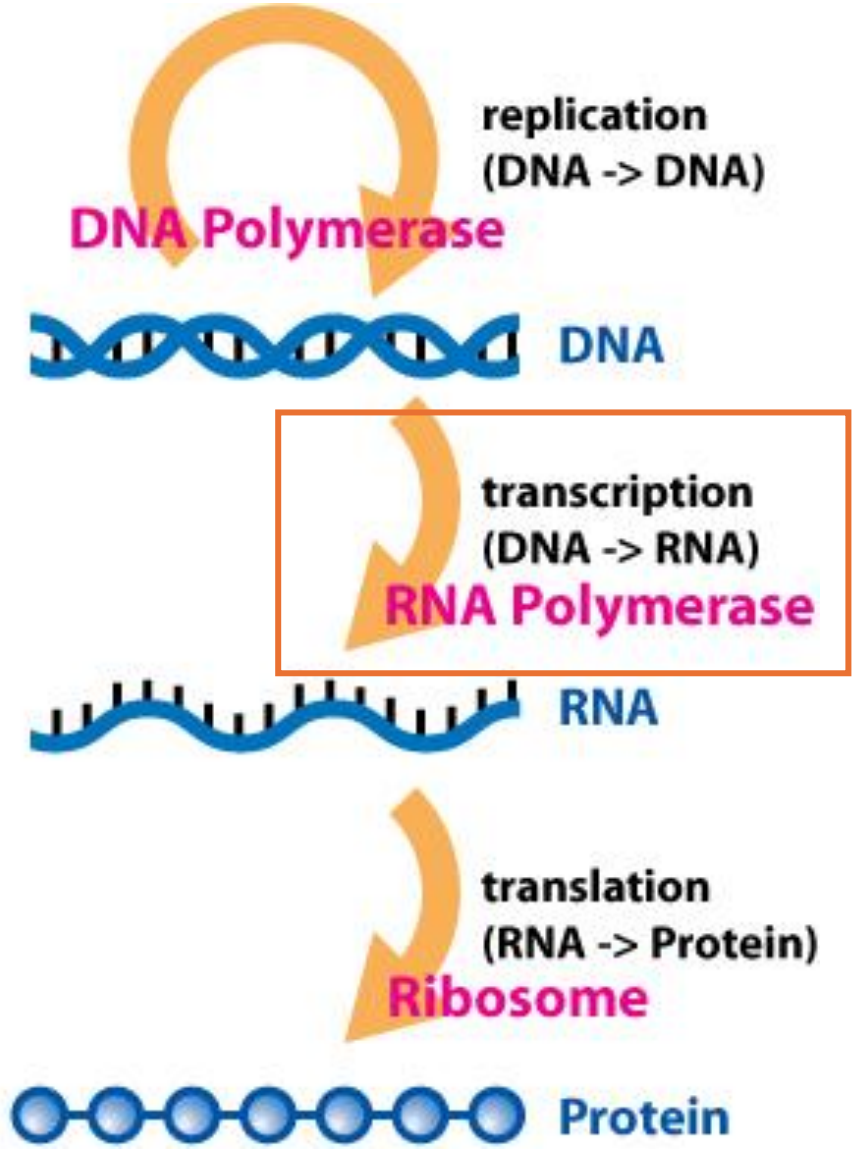
Siwei Qian

PhD Candidate, Landick lab

06/30/2026

Flow of genetic information within a biological system

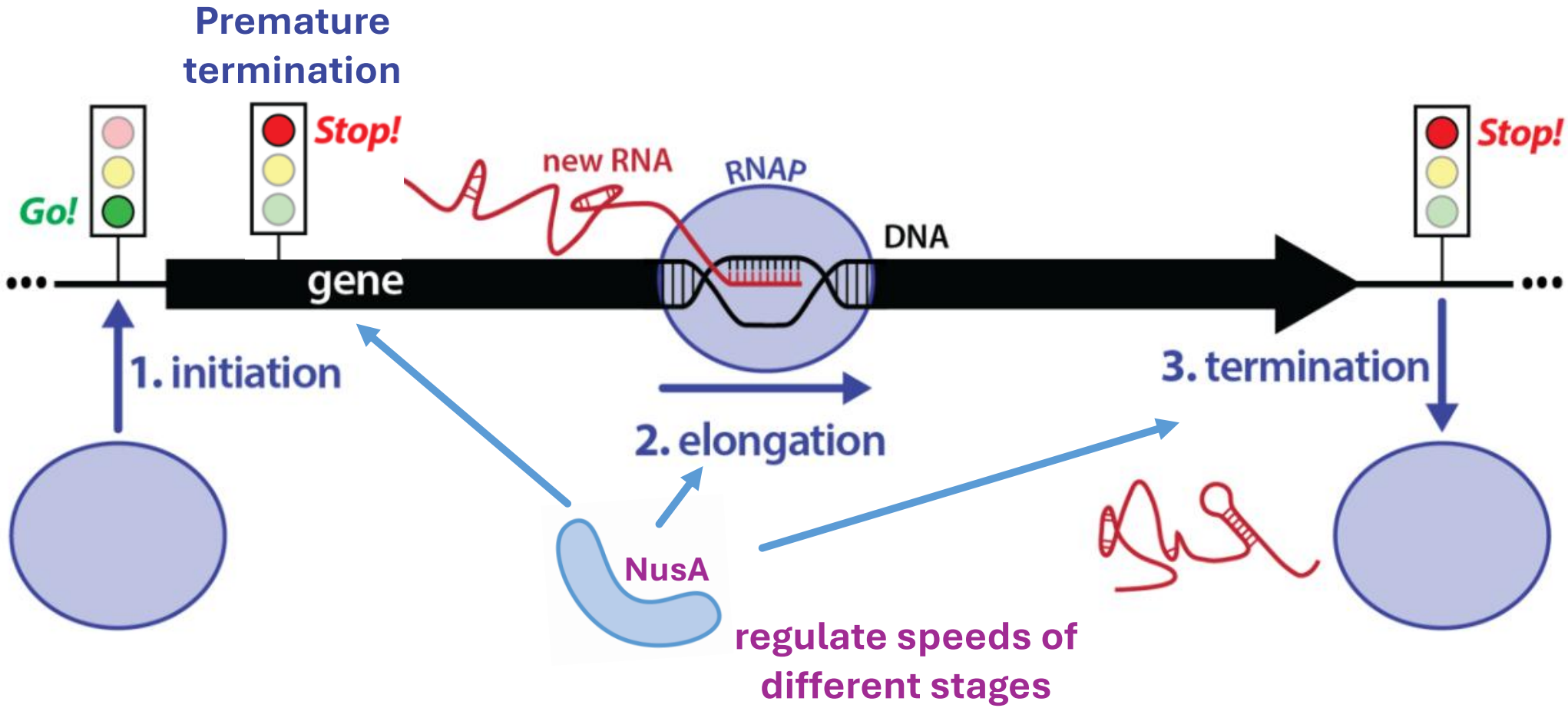
Central dogma



https://en.wikipedia.org/wiki/Central_dogma_of_molecular_biology

Transcription is the central hub for gene expression and its regulation

RNAP: RNA Polymerase

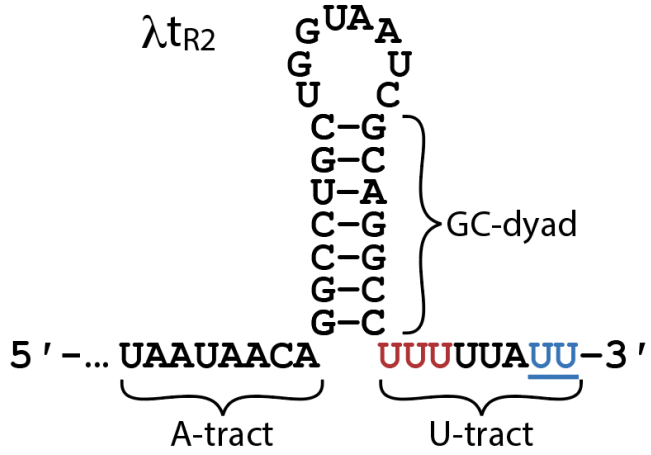


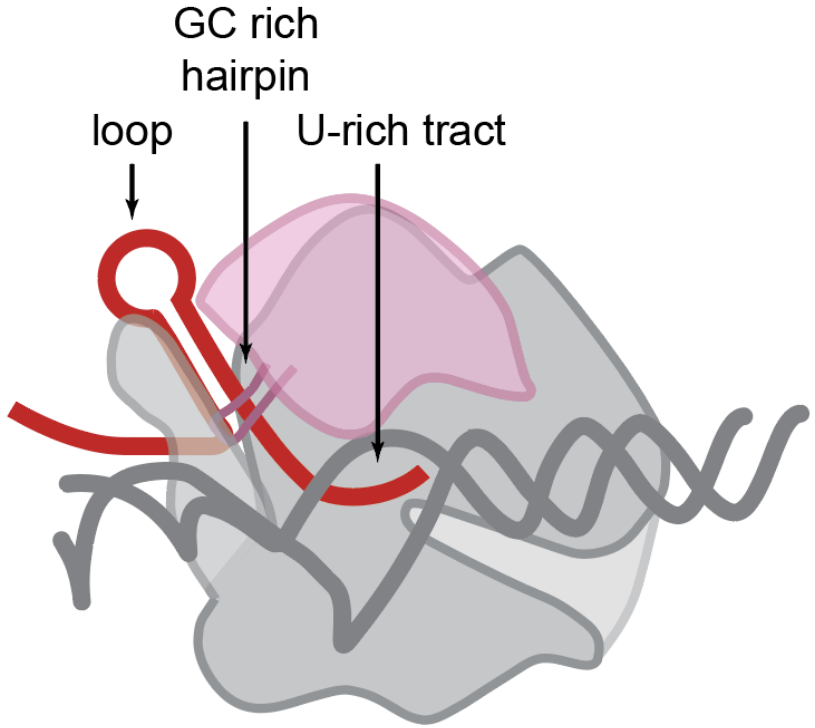
Ray-Soni, Ananya. Ph.D., The University of Wisconsin - Madison, 2017. ProQuest Dissertations & Theses Global (1993494005).

Intrinsic termination

- Depends on weak RNA–DNA hybrid and RNA structure formed by newly produced RNA (nascent RNA)
  - can be regulated by protein factors
- Intrinsic terminator, structure requirement

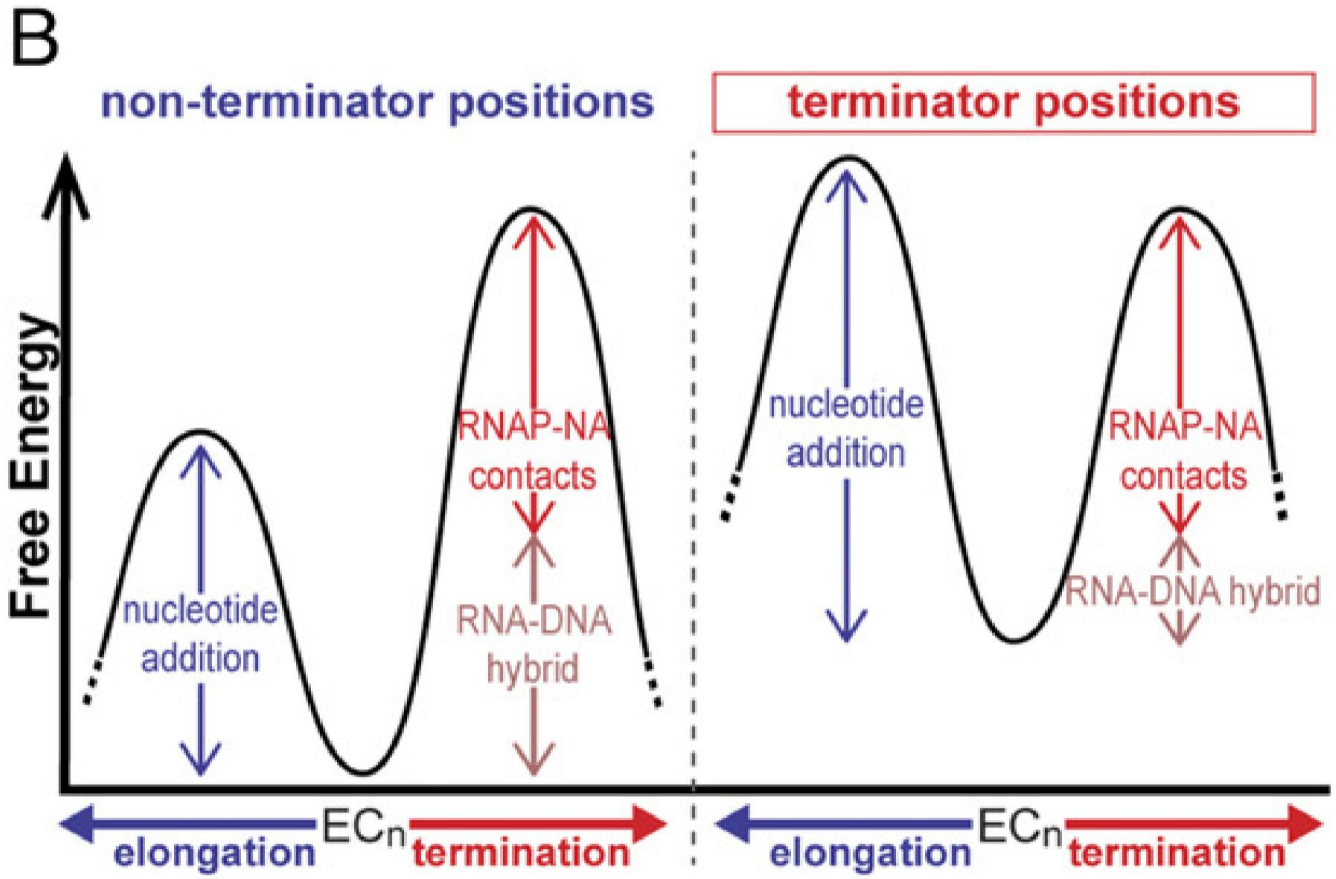
How does RNAP terminate during rapid and stable elongation?





Ray-Soni, Ananya, Rachel A. Mooney, and Robert Landick. *Proceedings of the National Academy of Sciences* 114, nc

Varying termination efficiency created from competition between intrinsic termination and elongation



TE determined by height of the activation energy

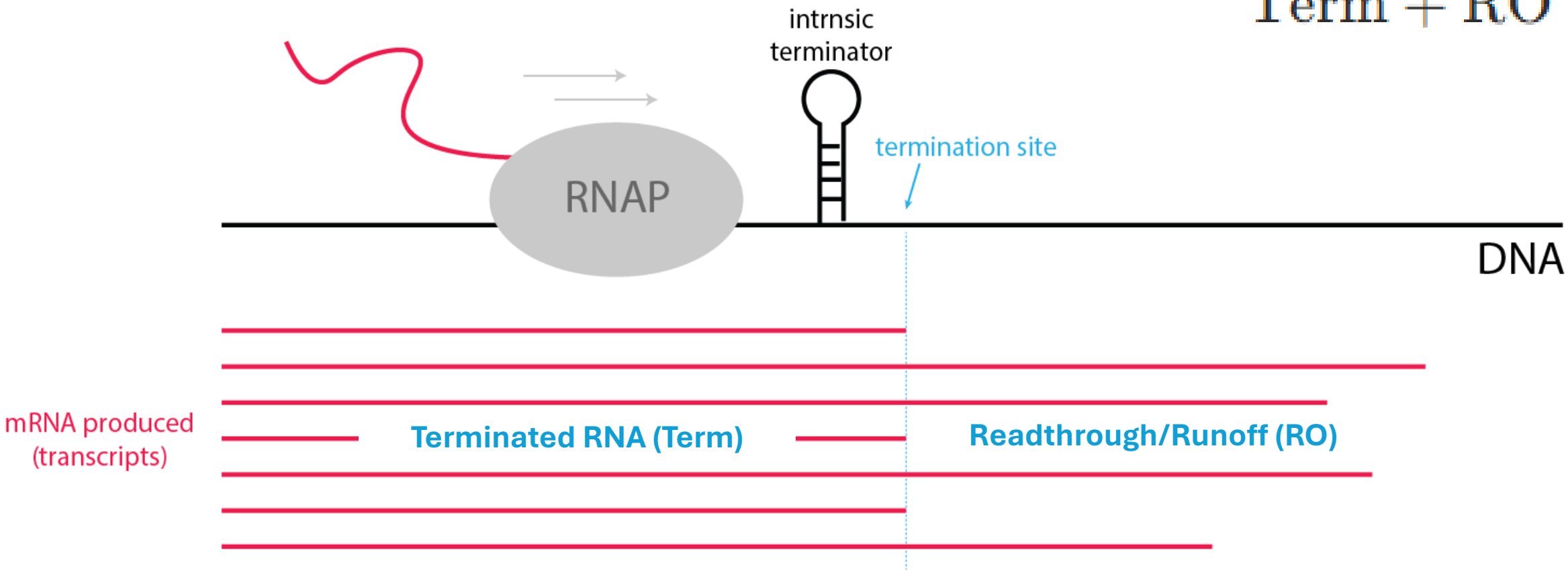
$$\mathrm{TE} = \frac{k_{\mathrm{release}}}{k_{\mathrm{release}} + k_{\mathrm{forward}}}$$

$$k = \kappa \frac{kT}{h} \exp\left(-\frac{\Delta G^{\ddagger}}{RT}\right)$$

Ray-Soni, et al. *Proceedings of the National Academy of Sciences* 114, no. 44 (2017)
Von hippel

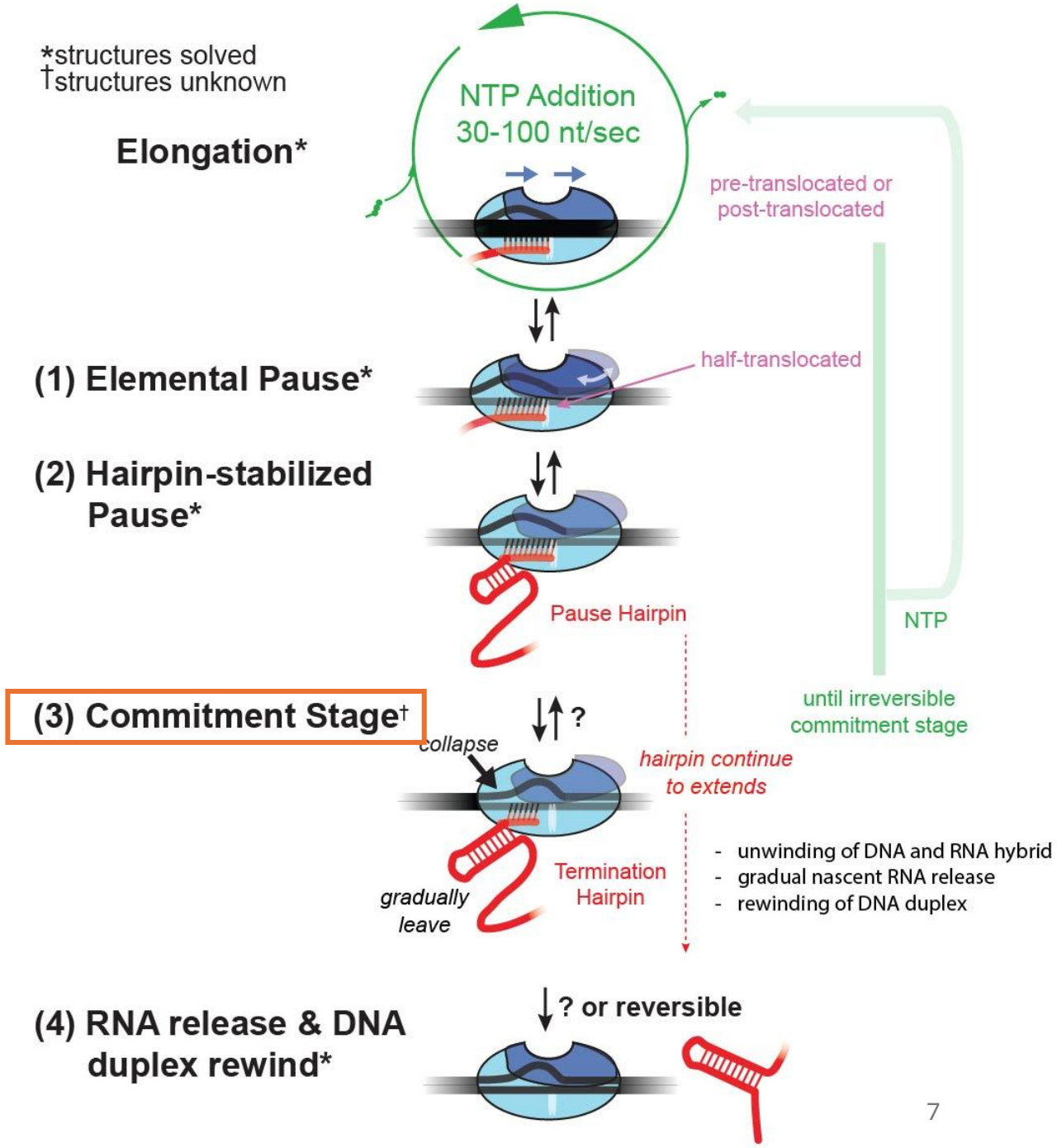
Termination Efficiency

$$TE = \frac{\text{Term}}{\text{Term} + \text{RO}}$$



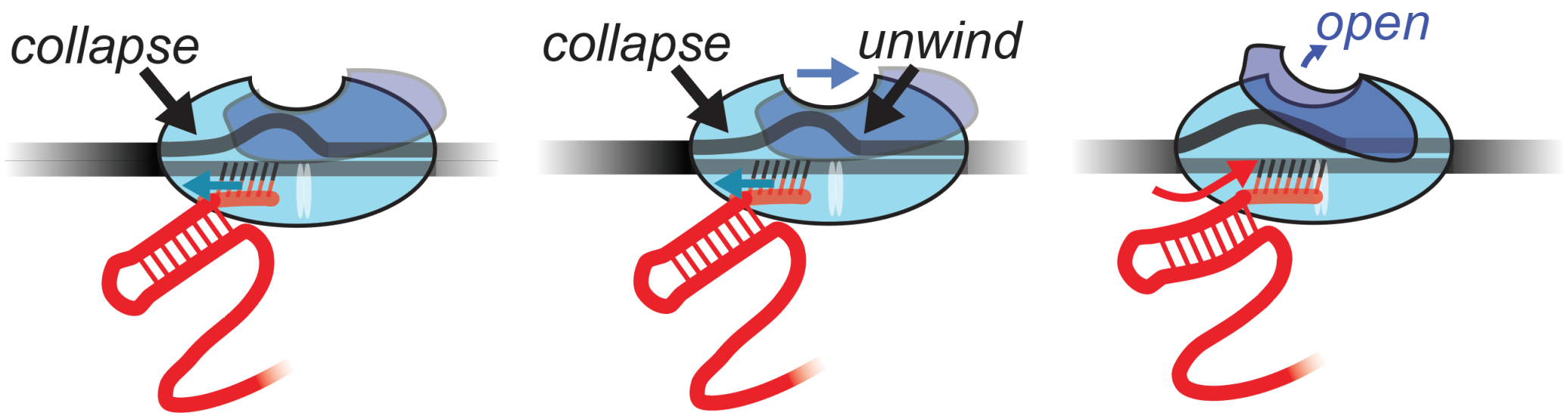
7 transcriptions happened on the DNA. 3 out of the 7 times, RNAP terminated at the terminator and 4 read-throughed. So the termination efficiency of this intrinsic terminator is $3 \div 7 \approx 0.429$

Current Model of Intrinsic termination mechanism



Wilson, Kevin S., and Peter H. von Hippel. no. 1 (1994): 36–51.
Peters, Jason M., et al. Journal of Molecular Biology, Mechanisms of Transcription, vol. 412, no. 5 (2011): 793–813.
You, Linlin, et al. Nature 613, no. 7945 (2023): 783–89.
Ray-Soni, et al. *Proceedings of the National Academy of Sciences* 114, no. 44 (2017)
Roberts, Jeffrey W. Journal of Molecular Biology, 60, vol. 431, no. 20 (2019): 4030–39.
Landick, Robert. Annual Review of Microbiology 75, no. 1 (2021): 291–314.

Hypothesized Mechanisms in the Commitment Stage



Hybrid Shearing

Forward Translocation

Hairpin Invasion

Depend on downstream U tract sequences
Depends on the energy of hairpin and energy of RNA-DNA hybrid

Not required to happen

Jeff Robert
Jason Peter
Ray Soni

Why need computational terminator design?

Basic Research

- Terminators aren't just endpoints — intragenic terminators, a regulation of gene expression
  - Example: control amino acid production (Trp operon), riboswitches
- We still don't understand them: the classic hairpin + U-tract model is incomplete — stem-preserving mutations can abolish termination (Cheng 1991)
- An interpretable sequence→strength model is a readout of mechanism — its features point to what physically governs termination

Designing intrinsic terminators -- Synthetic biology

- Gene Circuits need many strength-matched but sequence-diverse terminators
  - Gene expression output highly relate to the strength of intrinsic terminator at the end of gen
  - reuse >25 bp risks recombination, and repetitive DNA resists synthesis
- Prior thermodynamic models explain < half the variance (and only after excluding weak terminators) — not predictive enough to design to spec, or to generalize beyond E. coli

Battaglia, R. A., et al., bioRxiv 2026.05.29.728758 (2026)
Yanofsky, C. Nature 289 (1981): 751–758.

Intrinsic Termination: Features That Modulate TE

1. **RNA hairpin (H)**
2. **U-tract & RNA-DNA hybrid (U)**
3. **Bubble-edge sequences (B)**
4. **Downstream DNA ("front element")**
5. **Upstream / competing RNA structure**
6. **Kinetics: elongation rate, pausing, NTPs**
7. **Trans-acting protein factors**
8. **RNAP conformational determinants**
9. **Upstream transcription-bubble DNA (rewinding/collapse)**
10. **Organism-specific effects (mycobacteria)**

Types of models on intrinsic termination

- Classification: given genome sequence, predict location of intrinsic terminator
- Regression: given a candidate intrinsic terminator sequence, predict its strength/termination efficiency
- Generator: given TE and other features, generate a terminator

Nucleic Acid Destabilization & Kinetic Competition Model (von Hippel & Yager, 1991)

$$\Delta G^\circ_{\text{complex}} = \Delta G^\circ_{\text{DNA bubble}} + \Delta G^\circ_{\text{RNA-DNA hybrid}} + \Delta G^\circ_{\text{pol binding}}$$

$$G_T - G_E = \Delta G^\circ_{\text{RNA-DNA}}(n\text{-mer}) + \Delta G^\circ_{\text{RNA-RNA}}\,[(12-n)\text{-mer}] - \Delta G^\circ_{\text{RNA-DNA}}(12\text{-mer})$$

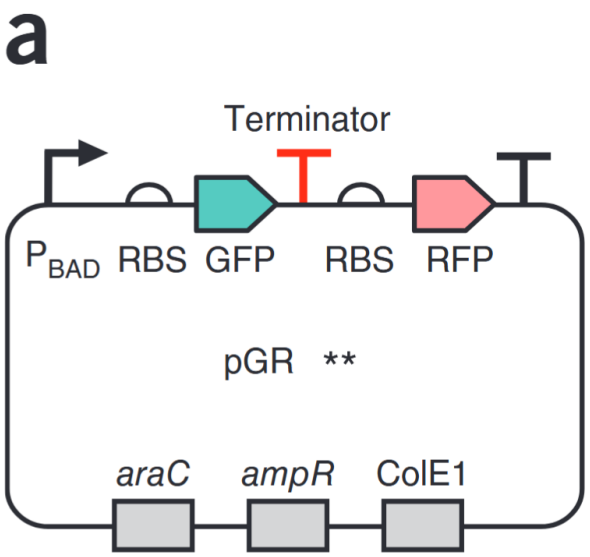
- Stability function acts as binary switch: drops from ~-13 to ~+1.2 kcal/mol over just 1-2 nt positions

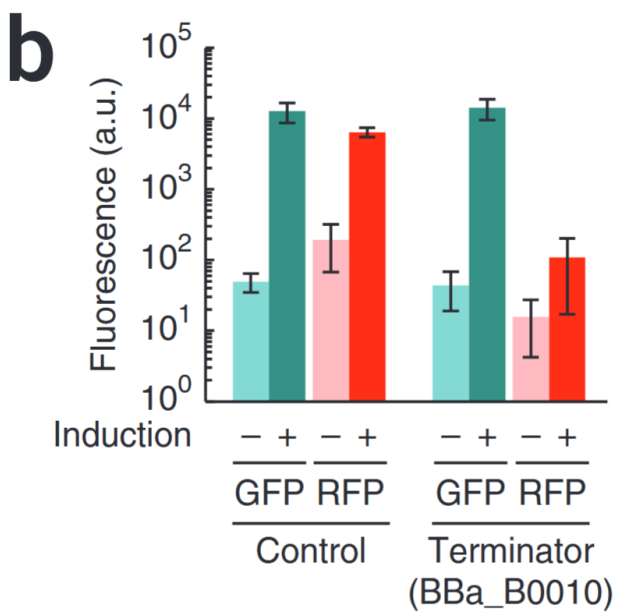
$$\text{TE} = \left[1 + \exp\left(-\frac{\Delta(\Delta G^\ddagger)}{RT}\right)\right]^{-1} \qquad \Delta(\Delta G^\ddagger) = \Delta G^\ddagger_{\text{forward}} - \Delta G^\ddagger_{\text{release}}$$

- Yields a sigmoid: only 2-4 kcal/mol change in relative barrier heights switches TE from <1% to >99%
- Key limitation: lumps multi-step processes into single rates; thermodynamic parameters for RNA-DNA hybrid were interpolated (no experimental data at the time)

Yager, T.D. & von Hippel, P.H. *Biochemistry* **30**, 1097–1118 (1991).
von Hippel, P.H. & Yager, T.D. *Proc. Natl. Acad. Sci. USA* **88**, 2307–2311 (1991).

In vivo fluorescence reporter





- Stronger terminator → less RFP relative to GFP
- T_S = (GFP_term / RFP_term) / (GFP_0 / RFP_0); TE = 1 - 1/T_S
- E. coli DH5α, LB, 10 mM L-arabinose, 3 hr induction, flow cytometry, ≥3 bio replicates, 30,000 events/sample
- Library: 227 natural + 90 reverse + 265 synthetic = 582 terminators
- Key limitation: conflates transcriptional termination with mRNA stability, cryptic promoters, RBS occlusion; no baseline correction (GFP/RFP positions never swapped)

Chen, Y.-J., Liu, P., Nielsen, A.A.K., Brophy, J.A.N., Clancy, K., Peterson, T. & Voigt, C.A. *Nat. Methods* **10**, 659–664 (2013).

Chen et al 2013 regression on TE model

- 4 free energy features, all from sequence via ViennaRNA/nearest-neighbor parameters:
  - ΔG_U (U-tract hybrid energy), ΔG_L (loop closure), ΔG_B (stem base stacking, bottom 3 bp), ΔG_A (A-tract extension energy)

$$T_S = 1 + \frac{1}{\underbrace{B_1\, e^{\beta_1 \Delta G_L}}_{\text{step 1: hairpin folding}} + \underbrace{B_4\, e^{\beta_4(\Delta G_B + \Delta G_A - \Delta G_U)}}_{\text{step 2: U-tract ratcheting}} \left(1 + B_1\, e^{\beta_1 \Delta G_L}\right)}$$

- Hairpin folding treated as irreversible
- Only 4 fit parameters (B_1, B_4, β_1, β_4)
- **R^2 = 0.40** on all 582 terminators
- Notable negative result: total hairpin folding energy ΔG_H showed no correlation with T_S; stem length and loop length also uncorrelated Underpredicts pheA-1 by 8-fold (pseudoknot, not captured by model)

Chen, Y.-J., Liu, P., Nielsen, A.A.K., Brophy, J.A.N., Clancy, K., Peterson, T. & Voigt, C.A. *Nat. Methods* **10**, 659–664 (2013).

Kundlatsch et al. 2026 (bioRxiv preprint) -- XGBoost

- Data: same Chen 582-terminator dataset
- 16 hand-crafted sequence features across 4 regions (A-tract, hairpin, loop, U-tract); deliberately avoided ΔG
  - Novel features: Shannon entropy and state-change count (# nucleotide transitions)
  - U-tract Shannon entropy = single most important feature; state-change count = second
  - "Initial GC count" at hairpin base outperformed Chen's fixed-3-bp ΔG_B
- Feature selection: incremental evaluation of >130 candidates by R^2 contribution
- Model: XGBoost (n_estimators=4000, lr=0.001, max_depth=5, subsample=0.65)
  - Discarded configs with train-test R^2 gap > 0.3 or train R^2 > 0.9
- Train R^2 = 0.897, Test R^2 = 0.652 (5-fold CV); train-test gap = 0.245 Also tested: voting ensemble (0.611), MLP (0.491), linear regression (0.363)
- SHAP: features largely orthogonal; U-tract dominates, A-tract second, hairpin less than expected

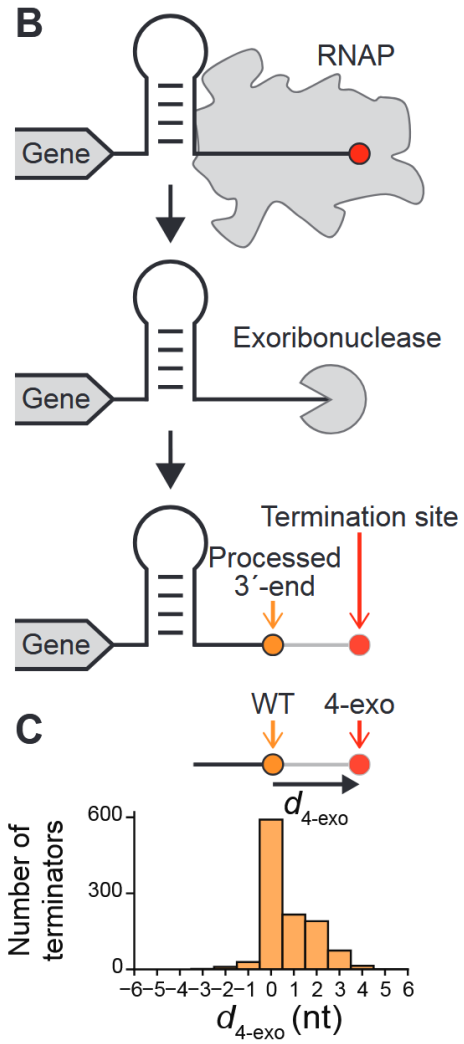
Kundlatsch, G.E., Neto, A.P.S., de Paiva, G.B., Rech, E., Duarte, L.T. & Pedrolli, D.B. *bioRxiv* (2026).
doi: 10.64898/2026.01.22.701115

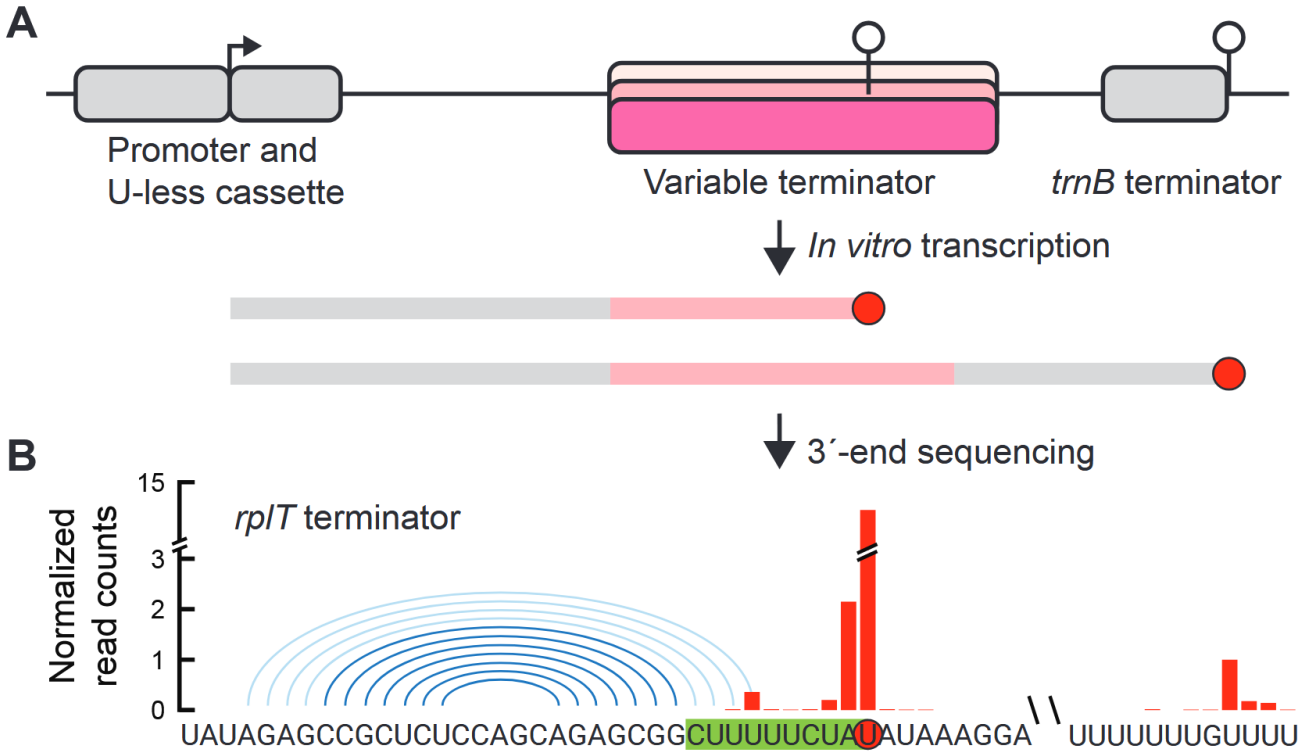
Li et al. 2025 (ACS Synth. Biol.) -- SVR + WGAN-GP

- Data: 606 E. coli terminators (Chen + RegulonDB + Dar & Sorek + Cambray)
- 252 combined features in 3 categories:
  - Sequence: 3-mer frequencies (64 features)
  - Thermodynamic: 7 ΔG values (ΔG_T, ΔG_U, ΔG_L, ΔG_H, ΔG_H/hairpin length, ΔG_H/stem length, ΔG_B)
  - Secondary structure region: 1-mer, 2-mer, C2SNAP per structural region + U_score (181 features)
- Feature selection: two-phase (rank by F-test, MI, RF importance → incremental addition tracking R^2)
  - Adding more features (4-mer, CKSNAP) actually decreased R^2 -- diminishing returns / overfitting
- Model: SVR (support vector regression)
- **Train R^2 = 0.7204, Test R^2 = 0.70** (held-out test set), RMSE = 1.3513
- Chen biophysical model on same test set: R^2 = 0.29
- Feature-to-sample ratio: 252:606 (relatively high)
- WGAN-GP for terminator generation: 18 tested experimentally, 72% had TE > 90%, matched BBa_B0010

Li, J., Wu, L.-F., Liu, K. & Ma, B.-G. *ACS Synth. Biol.* **14**, 3744–3752 (2025).

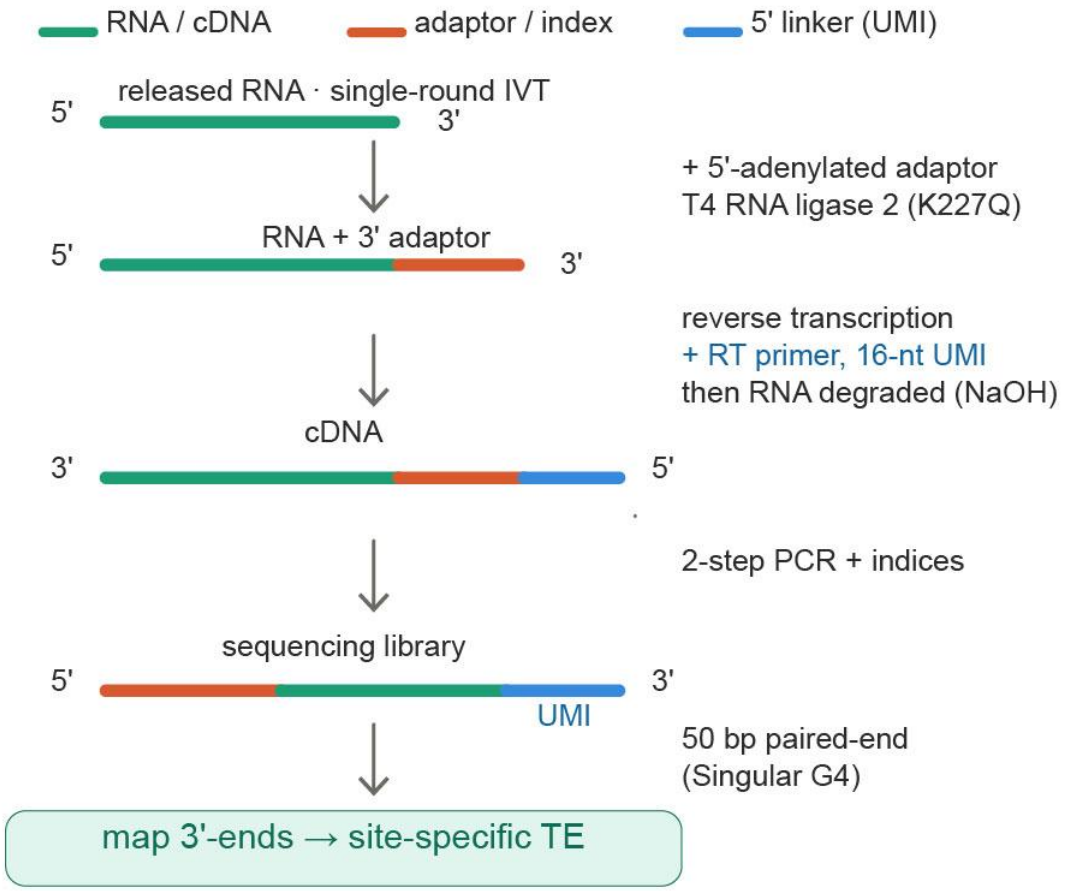
In vitro sequencing





Battaglia, R. A., et al., bioRxiv 2026.05.29.728758 (2026)

In vitro sequencing



only 24% of B. subtilis terminators produce identical 3' ends in vivo vs. in vitro (Battaglia et al. 2026)

Enables site-specific TE measurement (most terminators have multiple termination sites, each with different hairpin/U-tract context)

Battaglia, R. A., et al., bioRxiv 2026.05.29.728758 (2026)

Battaglia et al. 2026 (Gene-Wei Li lab, bioRxiv) -- HUB Model

- Data: thousands of B. subtilis + E. coli terminators, in vitro single-round transcription with species-matched RNAP
- Key discovery: phased sequence logos at strongest termination site revealed **bubble edge elements** that in vivo Term-seq could not resolve
  - Bubble edge resembles the elemental pause consensus sequence
  - Explains the λtR2 GC > CG mystery: G-C and C-G are thermodynamically near-identical for hairpin stability → the effect must be base identity/orientation at the bubble edge, not hairpin ΔG
- Mutational validation on diverse representative terminators confirmed bubble edge strongly influences TE
- Model: XGBoost with sequence positions encoded as unicode integers (each position = individual feature)
  - Features chosen from biology, not from computational feature-importance screening like Kundlatsch
  - No per-feature ablation / SHAP analysis (a gap vs. Kundlatsch)
- R^2 progression: **HU (classical) = 0.57 → HUB = 0.68 → All (full-range, -41 to +19) = 0.71**
  - Same trend across different transcription factor conditions and NTP concentrations
- Accuracy: 69% of HUB predictions within 2-fold of measured TE
- HUBfinder (genome-scale localization): in C. crescentus (67% GC genome it was never trained on), ~51% of HUBfinder calls within 10 nt of a mapped 3' end vs. ~9% for HU model → HUB is a universal feature, not species-specific
- Hairpin selection method is heuristic (not systematic)

Battaglia, R. A., et al., bioRxiv 2026.05.29.728758 (2026)

Summary of previous methods

Two axes of prior work:

- Data: in vivo noisy, in vitro = low throughput
- Model: white box/mechanistic vs black box/ML

Historically: white-box went with poor data; black-box (ML) went with better data but needs hand-fed features and is hard to interpret

Can we leverage the development of technology to get a more white box model?

What changed:

- high-throughput in vitro data on thousands of terminators (predicts on specific terminators library or natural terminators)
- better-measured free energies (RNA-DNA hybrid, DNA-DNA, RNA-RNA binding) and rates (single-molecule)
- improved terminator classification
- updated mechanistic understanding from biochemistry

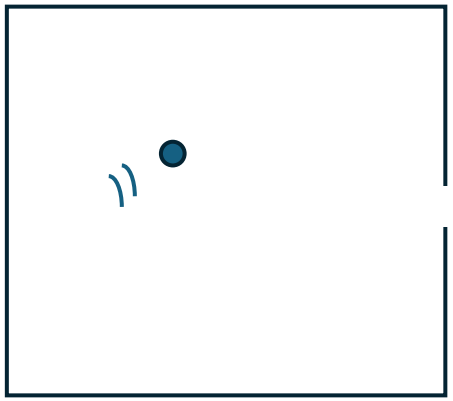
Define the task

Given a terminator sequence, predict its termination efficiency (TE)

Eventually: predict its sensitivity to conditions (NTP concentration, protein factors)

Approach: treat TE as the outcome of a stochastic kinetic competition, and compute it from sequence

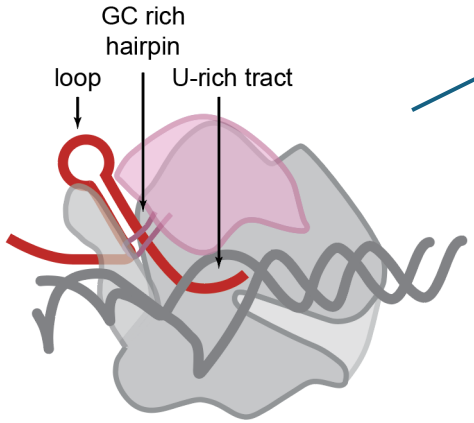
TE as a first-passage problem



Continuous state

Elongate

Release RNA



Discrete state

First passage/exit time:
when it exits?
with what probability?

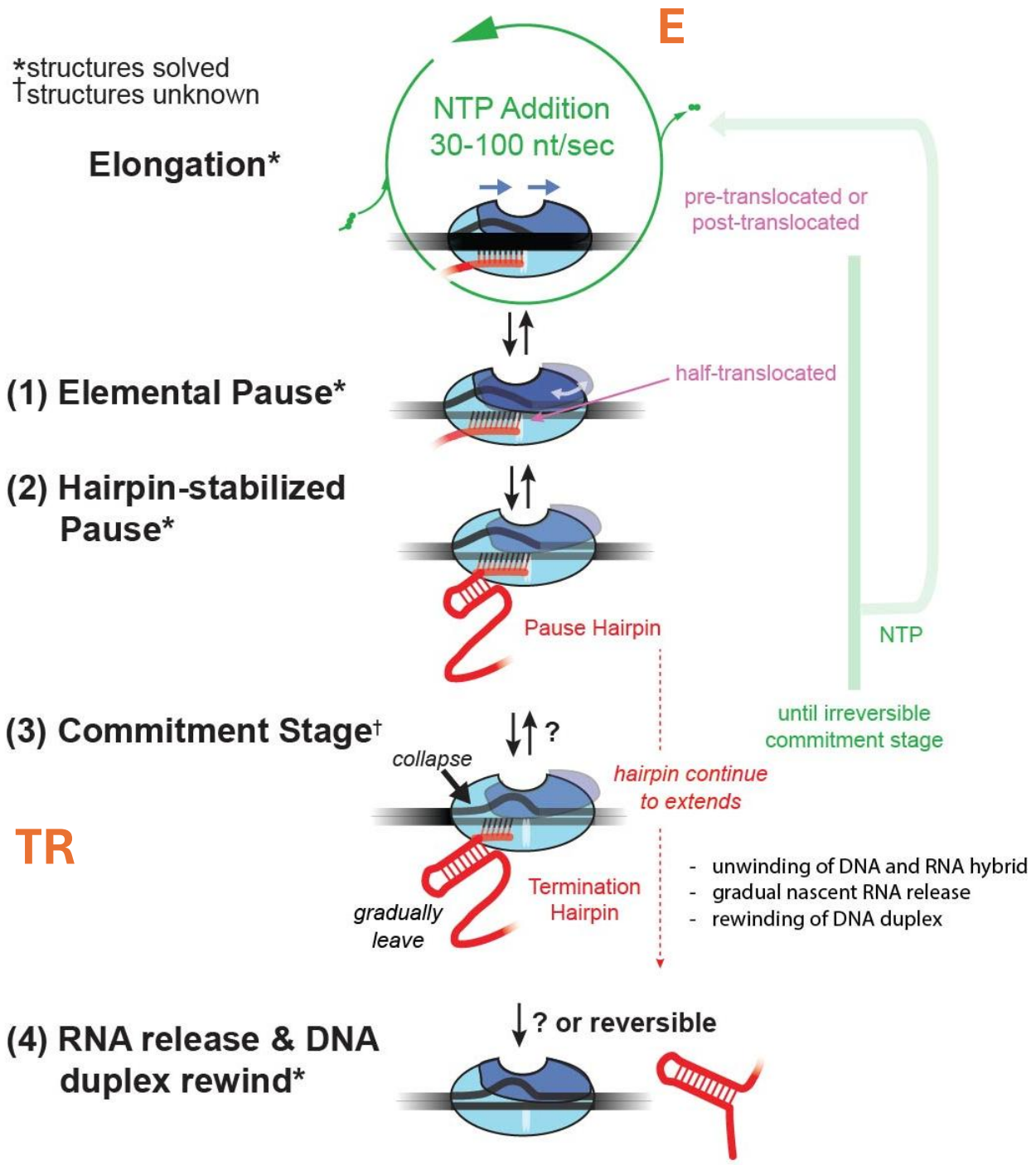
Absorbing state (End fate)
Transient state (still in play)

A 4-state kinetic model

$$E \xleftarrow{k_e} EC \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} TR \xrightarrow{k_2} T$$

- EC = paused elongation complex (the decision point)
- TR = termination-ready (hairpin nucleated, poised to commit)
- T = RNA released (absorbing: terminated)
- E = escaped back to processive elongation (absorbing: read-through)

- Four rate constants: k_1 (hairpin nucleation), k_-1 (fraying back), k_2 (commitment / hybrid disruption), k_e (elongation escape).



TE is a splitting probability

- Define h_i = probability of ending at T, given you start in state i
- Boundary values are forced:
  - $h_T = 1$ (already terminated),
  - $h_E = 0$ (already read through)
- TE = h_{EC} (start from the paused complex, probability you reach T)
- "Splitting": trajectories from EC split between the two fates; TE is the fraction reaching T, and read-through = 1 − TE

Solve by conditioning on the first jump

idea: from any state, condition on where the first jump goes.

Probability of jumping to a neighbor = its rate / total exit rate (competing exponential clocks). The process is memoryless, so after landing you just inherit that neighbor's h-value

$$h_{EC} = \frac{k_1}{k_1 + k_e} h_{TR} \qquad h_{TR} = \frac{k_2}{k_2 + k_{-1}} \cdot 1 + \frac{k_{-1}}{k_2 + k_{-1}} h_{EC}$$

Termination efficiency formula

$$TE = \frac{k_1 k_2}{k_1 k_2 + k_e(k_{-1} + k_2)}$$

- Only ratios of rates appear → TE is scale-free in time (the "which fate" question doesn't depend on absolute timescales)
- Sanity checks:
  - $k_e \to 0$ (no escape): TE → 1
  - $k_e \to \infty$ (instant escape): TE → 0
  - $k_{-1} \to 0$ (no fraying back): TE → $k_1 / (k_1 + k_e)$
  - $k_2 \to \infty$ (instant commitment): TE → $k_1 / (k_1 + k_e)$

From sequence to rates via free energy

- Each rate set by an energy barrier: $k = k_0 \cdot \exp(-\Delta G^{\ddagger} / kT)$
- k_1 (nucleation): barrier to forming the hairpin stem nucleus → ΔG_{stem}
- k_2 (commitment): governed by the competition between hybrid and stem (from single-molecule/optical-trap work) → $E_{stem} - E_{hybrid}$
- k_{-1} (fraying): reverse of nucleation → tied to ΔG_{stem} (fixed by detailed balance, not free)
- k_e (escape): set by NTP concentration and pause propensity → [NTP], pause score

Free energies are computed from sequence (nearest-neighbor of nucleic acid duplex); the pause term can be constrained directly from measured pause kinetics or calculate pause score